

BAYESIAN LATENT CLASS REGRESSION AND VARIABLE SELECTION WITH APPLICATIONS TO SLEEP PATTERNS DATA: SUPPLEMENTARY MATERIAL

CONTENTS

S1 Simulation Studies	1
S1.1 Simulation 1: Parameter Estimates and Selection Results	1
S1.2 Simulation 2: Parameter Estimates and Selection Results	4
S2 CSHQ Sleep Data Analysis	6
S2.1 Study Sample Characteristics	6
S2.2 Model Parameter Estimates	7
S2.3 Variable Selection Results	10
S2.4 Supplementary Visualisations	11
S3 Technical Derivations	13
S3.1 Derivation of Pólya-Gamma Sampling Method for LCR	13
References	14

S1. Simulation Studies.

S1.1. *Simulation 1: Parameter Estimates and Selection Results.* Table S1 contains the true and estimated values of the item probability parameters for the full LCR model for simulation 1, computed by posterior mean from Gibbs sampling. Table S2 contains the estimates for the item probability parameters from the item selection model computed post-hoc. Table S3 contains posterior inclusion probabilities for each item variable in simulation 1. Figure S1 contains a ridgeline plot illustrating the posterior distributions of the regression coefficients for the full LCR model without variable selection applied to simulation 1.

Var.	Cat.	True 1	Est. 1 (sd)	True 2	Est. 2 (sd)
1	1	0.15	0.08 (0.03)	0.70	0.74 (0.03)
	2	0.25	0.30 (0.03)	0.20	0.16 (0.02)
	3	0.60	0.61 (0.04)	0.10	0.10 (0.02)
2	1	0.20	0.23 (0.03)	0.55	0.55 (0.03)
	2	0.35	0.32 (0.03)	0.30	0.29 (0.03)
	3	0.45	0.45 (0.04)	0.15	0.15 (0.02)
3	1	0.10	0.14 (0.03)	0.80	0.82 (0.03)
	2	0.15	0.15 (0.03)	0.15	0.13 (0.02)
	3	0.75	0.71 (0.04)	0.05	0.05 (0.02)
4	1	0.25	0.26 (0.03)	0.45	0.44 (0.03)
	2	0.40	0.35 (0.04)	0.35	0.37 (0.03)
	3	0.35	0.40 (0.04)	0.20	0.19 (0.02)
5	1	0.40	0.39 (0.04)	0.40	0.44 (0.03)
	2	0.50	0.53 (0.04)	0.50	0.45 (0.03)
	3	0.10	0.08 (0.02)	0.10	0.11 (0.02)
6	1	0.70	0.75 (0.03)	0.70	0.72 (0.03)
	2	0.10	0.07 (0.02)	0.10	0.09 (0.02)
	3	0.20	0.18 (0.03)	0.20	0.19 (0.02)
7	1	0.10	0.08 (0.02)	0.10	0.09 (0.02)
	2	0.50	0.50 (0.04)	0.50	0.52 (0.03)
	3	0.40	0.42 (0.04)	0.40	0.38 (0.03)
8	1	0.33	0.28 (0.03)	0.33	0.32 (0.03)
	2	0.33	0.34 (0.03)	0.33	0.34 (0.03)
	3	0.33	0.38 (0.04)	0.33	0.34 (0.03)

TABLE S1

True and estimated values of the item probability parameters for the full LCR model for simulation 1.

Var.	Cat.	True 1	Est. 1 (sd)	True 2	Est. 2 (sd)
1	1	0.15	0.08 (0.03)	0.70	0.73 (0.03)
	2	0.25	0.30 (0.03)	0.20	0.16 (0.02)
	3	0.60	0.62 (0.04)	0.10	0.11 (0.02)
2	1	0.20	0.23 (0.03)	0.55	0.55 (0.03)
	2	0.35	0.32 (0.03)	0.30	0.29 (0.03)
	3	0.45	0.45 (0.04)	0.15	0.15 (0.02)
3	1	0.10	0.13 (0.03)	0.80	0.82 (0.03)
	2	0.15	0.15 (0.03)	0.15	0.13 (0.02)
	3	0.75	0.72 (0.04)	0.05	0.05 (0.02)
4	1	0.25	0.26 (0.03)	0.45	0.44 (0.03)
	2	0.40	0.35 (0.04)	0.35	0.37 (0.03)
	3	0.35	0.40 (0.04)	0.20	0.19 (0.02)
5	1	0.40	0.40 (0.04)	0.40	0.44 (0.03)
	2	0.50	0.52 (0.04)	0.50	0.46 (0.03)
	3	0.10	0.08 (0.02)	0.10	0.11 (0.02)
6	1	0.70	0.74 (0.04)	0.70	0.72 (0.03)
	2	0.10	0.07 (0.02)	0.10	0.09 (0.02)
	3	0.20	0.18 (0.03)	0.20	0.19 (0.02)
7	1	0.10	0.08 (0.02)	0.10	0.09 (0.02)
	2	0.50	0.50 (0.04)	0.50	0.52 (0.03)
	3	0.40	0.42 (0.04)	0.40	0.39 (0.03)
8	1	0.33	0.29 (0.03)	0.33	0.32 (0.03)
	2	0.33	0.34 (0.04)	0.33	0.34 (0.03)
	3	0.33	0.34 (0.04)	0.33	0.34 (0.03)

TABLE S2

Estimated values of the item probability parameters for the item selection model for simulation 1.

Variable	1	2	3	4	5	6	7	8
PIP (item)	1.00	1.00	1.00	1.00	0.08	0.02	0.03	0.04
PIP (simul)	1.00	1.00	1.00	1.00	0.07	0.02	0.03	0.04

TABLE S3

Posterior inclusion probabilities (PIP) for each item variable in the LCR model for simulation 1. PIP (item): item selection alone; PIP (simul): simultaneous selection.

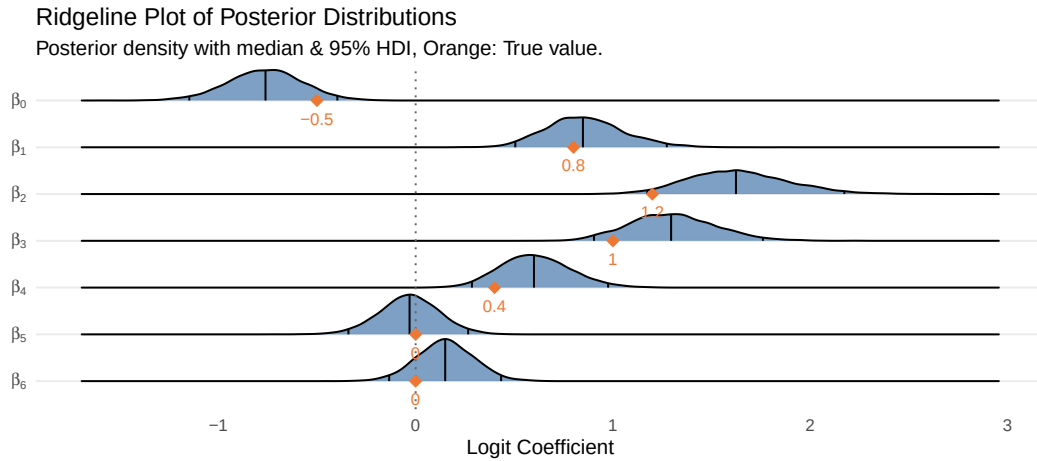


FIG S1. Ridgeline plots of posterior distribution of regression coefficients for simulation 1.

S1.2. Simulation 2: Parameter Estimates and Selection Results. Table S4 contains the true and estimated values of the item probability parameters for the collapsed LCR model for simulation 2, computed post-hoc. Table S5 contains posterior inclusion probabilities for each item variable in simulation 2. Figure S2 contains a ridgeline plot illustrating the posterior distributions of the regression coefficients for the LCR model with item selection only applied to simulation 2.

Var.	Cat.	True 1	Est. 1 (sd)	True 2	Est. 2 (sd)	True 3	Est. 3 (sd)
1	1	0.15	0.18 (0.03)	0.60	0.58 (0.04)	0.80	0.80 (0.05)
	2	0.85	0.82 (0.03)	0.40	0.42 (0.04)	0.20	0.20 (0.05)
2	1	0.25	0.24 (0.03)	0.45	0.45 (0.04)	0.70	0.77 (0.06)
	2	0.75	0.76 (0.03)	0.55	0.55 (0.04)	0.30	0.23 (0.06)
3	1	0.70	0.70 (0.04)	0.20	0.23 (0.03)	0.65	0.71 (0.08)
	2	0.30	0.30 (0.04)	0.80	0.77 (0.03)	0.35	0.29 (0.08)
4	1	0.10	0.08 (0.02)	0.35	0.39 (0.04)	0.70	0.64 (0.07)
	2	0.25	0.28 (0.03)	0.40	0.34 (0.04)	0.20	0.26 (0.06)
	3	0.65	0.64 (0.04)	0.25	0.27 (0.03)	0.10	0.09 (0.04)
5	1	0.20	0.18 (0.03)	0.25	0.29 (0.04)	0.65	0.66 (0.07)
	2	0.15	0.17 (0.03)	0.60	0.59 (0.04)	0.25	0.25 (0.06)
	3	0.65	0.65 (0.04)	0.15	0.12 (0.03)	0.10	0.09 (0.04)
6	1	0.15	0.11 (0.03)	0.50	0.45 (0.03)	0.75	0.79 (0.06)
	2	0.20	0.21 (0.03)	0.35	0.38 (0.04)	0.15	0.13 (0.05)
	3	0.65	0.68 (0.04)	0.15	0.16 (0.03)	0.10	0.09 (0.03)
7	1	0.10	0.10 (0.02)	0.25	0.25 (0.03)	0.60	0.51 (0.06)
	2	0.15	0.13 (0.03)	0.35	0.33 (0.03)	0.25	0.28 (0.06)
	3	0.25	0.23 (0.03)	0.25	0.25 (0.03)	0.10	0.15 (0.04)
	4	0.50	0.53 (0.04)	0.15	0.17 (0.03)	0.05	0.06 (0.03)
8	1	0.15	0.20 (0.03)	0.20	0.20 (0.03)	0.55	0.67 (0.07)
	2	0.20	0.14 (0.03)	0.45	0.45 (0.04)	0.20	0.16 (0.06)
	3	0.20	0.20 (0.03)	0.25	0.26 (0.03)	0.15	0.12 (0.04)
	4	0.45	0.46 (0.04)	0.10	0.10 (0.02)	0.10	0.05 (0.03)
9	1	0.40	0.35 (0.04)	0.40	0.43 (0.04)	0.40	0.39 (0.07)
	2	0.50	0.53 (0.04)	0.50	0.46 (0.04)	0.50	0.47 (0.07)
	3	0.10	0.12 (0.02)	0.10	0.11 (0.02)	0.10	0.14 (0.04)
10	1	0.70	0.72 (0.04)	0.70	0.72 (0.04)	0.70	0.68 (0.08)
	2	0.10	0.10 (0.02)	0.10	0.11 (0.02)	0.10	0.15 (0.04)
	3	0.20	0.18 (0.03)	0.20	0.17 (0.03)	0.20	0.17 (0.05)
11	1	0.20	0.21 (0.03)	0.20	0.20 (0.03)	0.20	0.20 (0.05)
	2	0.20	0.18 (0.03)	0.20	0.20 (0.03)	0.20	0.15 (0.04)
	3	0.20	0.22 (0.03)	0.20	0.20 (0.03)	0.20	0.23 (0.05)
	4	0.20	0.21 (0.03)	0.20	0.20 (0.03)	0.20	0.20 (0.05)
	5	0.20	0.19 (0.03)	0.20	0.20 (0.03)	0.20	0.22 (0.05)
12	1	0.10	0.15 (0.03)	0.10	0.11 (0.02)	0.10	0.08 (0.03)
	2	0.15	0.17 (0.03)	0.15	0.13 (0.02)	0.15	0.16 (0.04)
	3	0.20	0.16 (0.03)	0.20	0.20 (0.03)	0.20	0.25 (0.05)
	4	0.25	0.24 (0.03)	0.25	0.21 (0.03)	0.25	0.21 (0.05)
	5	0.30	0.28 (0.03)	0.30	0.34 (0.03)	0.30	0.31 (0.06)
13	1	0.20	0.20 (0.03)	0.20	0.20 (0.03)	0.20	0.23 (0.05)
	2	0.30	0.27 (0.03)	0.30	0.27 (0.03)	0.30	0.33 (0.06)
	3	0.30	0.31 (0.04)	0.30	0.30 (0.03)	0.30	0.29 (0.06)
	4	0.10	0.10 (0.02)	0.10	0.14 (0.03)	0.10	0.09 (0.03)
	5	0.10	0.11 (0.02)	0.10	0.09 (0.02)	0.10	0.06 (0.03)

TABLE S4

True and estimated values of the item probability parameters for the item selection model for dataset 2.

Variable	1	2	3	4	5	6	7	8	9	10	11	12	13
PIP (item)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.01	0.00	0.00	0.00	0.00
PIP (simul)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.01	0.00	0.00	0.00	0.00

TABLE S5

Posterior inclusion probabilities (PIP) for each item variable in the LCR model for dataset 2. PIP (item): item selection alone; PIP (simul): simultaneous selection.

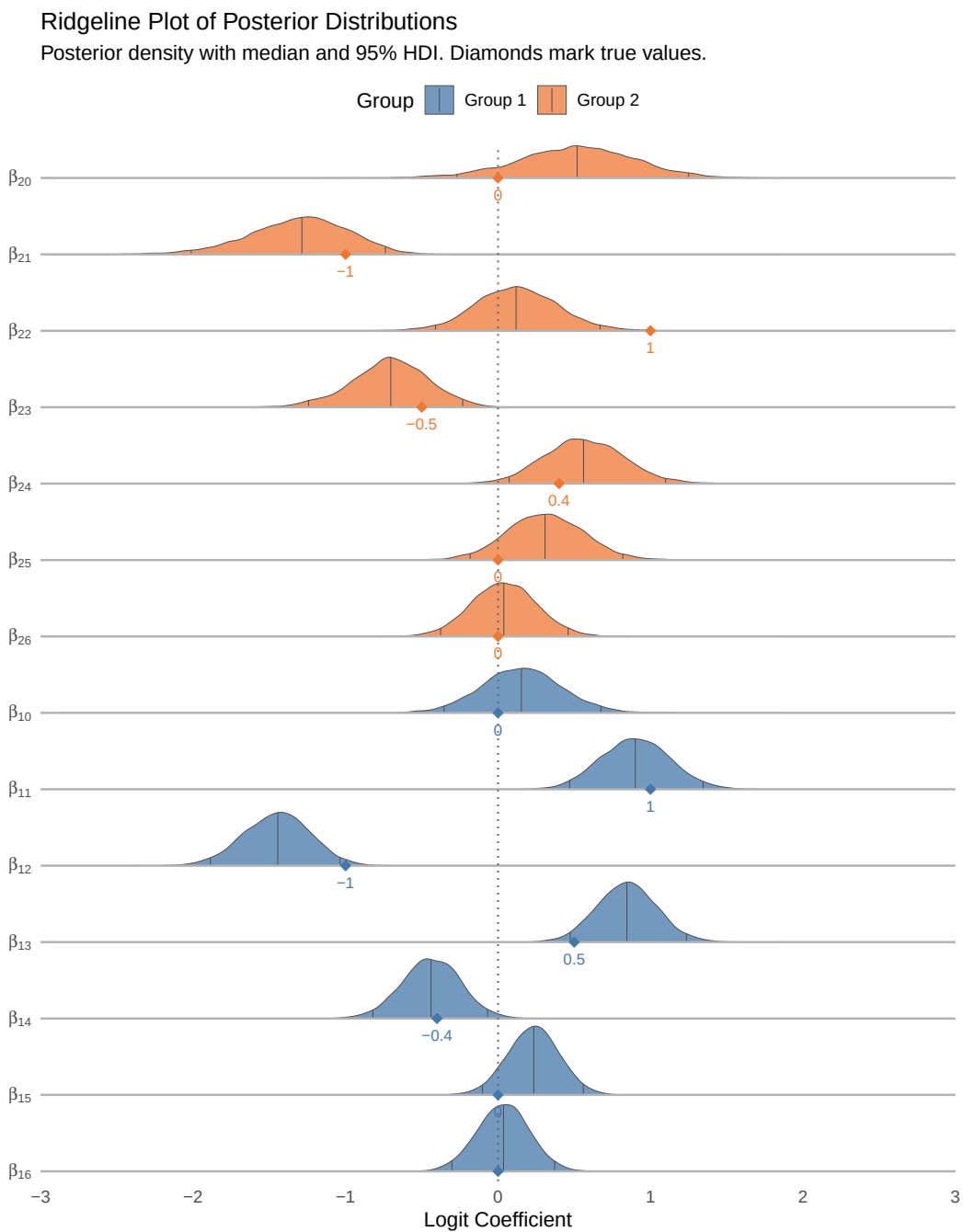


FIG S2. Ridgeline plots of posterior distribution of regression coefficients for simulation 2.

S2. CSHQ Sleep Data Analysis.

S2.1. *Study Sample Characteristics.* Table S6 contains demographic and diagnostic information for the sleep patterns dataset.

Demographic		<i>n</i>	%
Gender	Female	69	46.6
	Male	79	53.4
Diagnosis	Neurotypical	104	70.3
	Neurodiverse	44	29.7
Diagnosis/comorbidity (ND)		<i>n</i>	% (ND group)
ASD		19	43.18
ID/ASD		6	13.64
ADHD/ASD		5	11.36
Dyslexia		3	6.82
ADHD		2	4.55
ASD/ADHD/Dyslexia		1	2.27
ID/ASD/ADHD		1	2.27
ASD/GDD		1	2.27
ADHD/Dyslexia		1	2.27
ID/Dyslexia		1	2.27
ID		1	2.27
Dyspraxia		1	2.27
Tourette's		1	2.27
Brain Abnormality		1	2.27

TABLE S6

Demographic and diagnostic summary information for the CSHQ dataset (ND = neurodiverse, ASD = Autism Spectrum Disorder, ID = Intellectual Disability, GDD = Global Developmental Delay)

S2.2. Model Parameter Estimates. Table S7 contains estimates for the item probability parameters computed post-hoc for the sleep data LCR model with item selection. Table S8 contains posterior mean values for LCR regression coefficient parameters, along with sd and 95% HDI values. Table S9 contains the estimates for the regression coefficient parameters for the ASD diagnosis variable with predictor selection.

Table S7: Estimated values of the item probability parameters for the item selection model for the CSHQ sleep data

Variable	Category		Est. 1 (sd)	Est. 2 (sd)	Est. 3 (sd)	Est. 4 (sd)
Bedtime Resistance	1	1	0.78 (0.06)	0.77 (0.09)	0.63 (0.10)	0.68 (0.11)
		2	0.18 (0.05)	0.18 (0.07)	0.30 (0.09)	0.27 (0.09)
		3	0.04 (0.03)	0.05 (0.04)	0.07 (0.05)	0.04 (0.04)
	2	1	0.94 (0.03)	0.75 (0.07)	0.90 (0.05)	0.51 (0.11)
		2	0.02 (0.02)	0.09 (0.05)	0.07 (0.04)	0.33 (0.10)
		3	0.05 (0.03)	0.16 (0.06)	0.03 (0.03)	0.16 (0.07)
	3	1	0.94 (0.03)	0.75 (0.07)	0.87 (0.06)	0.43 (0.11)
		2	0.02 (0.02)	0.07 (0.04)	0.09 (0.05)	0.38 (0.10)
		3	0.05 (0.03)	0.18 (0.07)	0.03 (0.03)	0.19 (0.08)
Sleep Onset Delay	5	1	0.89 (0.04)	0.14 (0.07)	0.77 (0.08)	0.06 (0.04)
		2	0.09 (0.04)	0.38 (0.09)	0.20 (0.08)	0.25 (0.09)
		3	0.02 (0.02)	0.47 (0.08)	0.04 (0.04)	0.69 (0.10)
	1		0.58 (0.06)	0.48 (0.09)	0.15 (0.07)	0.20 (0.09)

Continued on next page

Table S7 – *Continued from previous page*

Variable		Category	Est. 1 (sd)	Est. 2 (sd)	Est. 3 (sd)	Est. 4 (sd)
Sleep Duration	1	2	0.27 (0.06)	0.41 (0.09)	0.57 (0.09)	0.35 (0.10)
		3	0.15 (0.05)	0.11 (0.05)	0.28 (0.08)	0.44 (0.10)
		1	0.85 (0.05)	0.65 (0.09)	0.14 (0.07)	0.08 (0.06)
	2	2	0.12 (0.05)	0.29 (0.09)	0.55 (0.09)	0.57 (0.11)
		3	0.03 (0.02)	0.06 (0.04)	0.32 (0.09)	0.35 (0.10)
		1	0.86 (0.05)	0.65 (0.09)	0.15 (0.07)	0.05 (0.04)
	3	2	0.10 (0.04)	0.27 (0.09)	0.53 (0.09)	0.61 (0.10)
		3	0.05 (0.03)	0.09 (0.05)	0.32 (0.09)	0.35 (0.10)
		1	0.86 (0.05)	0.74 (0.08)	0.22 (0.08)	0.29 (0.09)
Sleep Anxiety	1	2	0.11 (0.04)	0.23 (0.08)	0.54 (0.09)	0.37 (0.10)
		3	0.03 (0.02)	0.03 (0.03)	0.24 (0.08)	0.33 (0.10)
		1	0.80 (0.05)	0.14 (0.07)	0.89 (0.06)	0.09 (0.06)
	2	2	0.13 (0.04)	0.22 (0.07)	0.06 (0.05)	0.27 (0.05)
		3	0.06 (0.03)	0.63 (0.09)	0.05 (0.04)	0.64 (0.04)
		1	0.70 (0.06)	0.35 (0.08)	0.67 (0.09)	0.31 (0.09)
	3	2	0.20 (0.05)	0.19 (0.07)	0.12 (0.06)	0.36 (0.10)
		3	0.10 (0.04)	0.46 (0.09)	0.21 (0.08)	0.32 (0.10)
		1	0.89 (0.04)	0.11 (0.06)	0.76 (0.08)	0.06 (0.04)
Night Waking	1	2	0.08 (0.04)	0.35 (0.08)	0.20 (0.08)	0.35 (0.10)
		3	0.02 (0.02)	0.54 (0.09)	0.04 (0.04)	0.59 (0.10)
		1	0.70 (0.06)	0.63 (0.08)	0.41 (0.09)	0.14 (0.08)
	2	2	0.27 (0.06)	0.26 (0.07)	0.40 (0.09)	0.47 (0.10)
		3	0.03 (0.02)	0.11 (0.05)	0.19 (0.07)	0.39 (0.10)
		1	0.90 (0.04)	0.48 (0.09)	0.71 (0.09)	0.18 (0.08)
	3	2	0.08 (0.04)	0.34 (0.08)	0.20 (0.08)	0.34 (0.10)
		3	0.02 (0.02)	0.17 (0.07)	0.10 (0.07)	0.48 (0.10)
		1	0.76 (0.06)	0.33 (0.08)	0.31 (0.08)	0.10 (0.06)
Parasomnias	1	2	0.21 (0.06)	0.46 (0.09)	0.33 (0.09)	0.27 (0.09)
		3	0.03 (0.02)	0.21 (0.07)	0.35 (0.09)	0.63 (0.10)
		1	0.94 (0.03)	0.64 (0.08)	0.41 (0.10)	0.36 (0.10)
	2	2	0.04 (0.03)	0.33 (0.08)	0.25 (0.08)	0.39 (0.10)
		3	0.02 (0.02)	0.03 (0.03)	0.33 (0.09)	0.25 (0.09)
		1	0.91 (0.06)	0.87 (0.09)	0.81 (0.10)	0.72 (0.11)
	3	2	0.05 (0.03)	0.08 (0.04)	0.09 (0.05)	0.12 (0.07)
		3	0.05 (0.03)	0.05 (0.04)	0.09 (0.05)	0.16 (0.07)
		1	0.68 (0.06)	0.51 (0.09)	0.77 (0.09)	0.43 (0.10)
	1	2	0.26 (0.06)	0.46 (0.09)	0.13 (0.07)	0.49 (0.11)
		3	0.05 (0.03)	0.03 (0.03)	0.10 (0.06)	0.08 (0.06)
		1	0.48 (0.06)	0.28 (0.08)	0.27 (0.08)	0.16 (0.08)
	2	2	0.43 (0.06)	0.54 (0.08)	0.35 (0.09)	0.34 (0.10)
		3	0.10 (0.04)	0.19 (0.07)	0.38 (0.09)	0.50 (0.10)
		1	0.95 (0.05)	0.91 (0.09)	0.88 (0.10)	0.83 (0.10)
	3	2	0.05 (0.03)	0.09 (0.05)	0.12 (0.06)	0.17 (0.08)
		1	0.89 (0.04)	0.67 (0.09)	0.75 (0.08)	0.29 (0.09)
		2	0.06 (0.03)	0.18 (0.07)	0.16 (0.07)	0.41 (0.10)
	5	3	0.06 (0.03)	0.16 (0.06)	0.09 (0.06)	0.30 (0.09)

Continued on next page

Table S7 – *Continued from previous page*

Variable		Category	Est. 1 (sd)	Est. 2 (sd)	Est. 3 (sd)	Est. 4 (sd)
Sleep Disordered Breathing	6	1	0.93 (0.05)	0.90 (0.08)	0.79 (0.10)	0.73 (0.11)
		2	0.07 (0.03)	0.10 (0.05)	0.21 (0.08)	0.27 (0.09)
	7	1	0.80 (0.06)	0.69 (0.09)	0.68 (0.10)	0.60 (0.11)
		2	0.19 (0.05)	0.26 (0.08)	0.26 (0.08)	0.35 (0.10)
		3	0.02 (0.02)	0.05 (0.04)	0.06 (0.04)	0.04 (0.04)
	1	1	0.82 (0.06)	0.72 (0.10)	0.72 (0.10)	0.62 (0.11)
		2	0.14 (0.04)	0.22 (0.07)	0.16 (0.07)	0.34 (0.10)
		3	0.05 (0.03)	0.05 (0.04)	0.12 (0.06)	0.04 (0.04)
	2	1	0.98 (0.05)	0.97 (0.08)	0.90 (0.09)	0.83 (0.10)
		2	0.02 (0.02)	0.03 (0.03)	0.10 (0.05)	0.17 (0.08)
	3	1	0.92 (0.05)	0.94 (0.08)	0.79 (0.10)	0.80 (0.11)
		2	0.08 (0.04)	0.06 (0.04)	0.21 (0.08)	0.20 (0.08)
Daytime Sleepiness	1	1	0.45 (0.07)	0.66 (0.09)	0.46 (0.10)	0.53 (0.11)
		2	0.35 (0.06)	0.16 (0.07)	0.43 (0.10)	0.30 (0.10)
		3	0.20 (0.05)	0.18 (0.07)	0.11 (0.06)	0.17 (0.08)
	2	1	0.68 (0.06)	0.77 (0.08)	0.16 (0.07)	0.22 (0.09)
		2	0.28 (0.06)	0.20 (0.08)	0.63 (0.09)	0.53 (0.10)
		3	0.03 (0.02)	0.03 (0.03)	0.21 (0.07)	0.25 (0.09)
	3	1	0.59 (0.07)	0.75 (0.09)	0.30 (0.09)	0.34 (0.10)
		2	0.33 (0.06)	0.22 (0.08)	0.34 (0.09)	0.44 (0.10)
		3	0.08 (0.04)	0.03 (0.03)	0.36 (0.07)	0.21 (0.08)
	4	1	0.59 (0.06)	0.75 (0.08)	0.30 (0.08)	0.34 (0.10)
		2	0.37 (0.06)	0.22 (0.08)	0.34 (0.09)	0.44 (0.10)
		3	0.03 (0.02)	0.03 (0.03)	0.36 (0.09)	0.21 (0.08)
	5	1	0.84 (0.05)	0.81 (0.07)	0.42 (0.09)	0.33 (0.10)
		2	0.14 (0.05)	0.16 (0.06)	0.46 (0.09)	0.55 (0.11)
		3	0.02 (0.02)	0.03 (0.03)	0.12 (0.06)	0.12 (0.07)
	6	1	0.62 (0.06)	0.60 (0.09)	0.16 (0.07)	0.09 (0.06)
		2	0.36 (0.06)	0.38 (0.09)	0.51 (0.10)	0.59 (0.10)
		3	0.02 (0.02)	0.03 (0.03)	0.33 (0.09)	0.31 (0.10)
	7	1	0.92 (0.06)	0.90 (0.09)	0.78 (0.10)	0.80 (0.11)
		2	0.05 (0.03)	0.05 (0.04)	0.12 (0.06)	0.04 (0.04)
		3	0.03 (0.02)	0.05 (0.04)	0.09 (0.05)	0.16 (0.07)
	8	1	0.82 (0.06)	0.77 (0.09)	0.76 (0.10)	0.53 (0.11)
		2	0.17 (0.05)	0.17 (0.07)	0.21 (0.08)	0.36 (0.10)
		3	0.02 (0.02)	0.05 (0.04)	0.03 (0.03)	0.12 (0.06)

Group	Parameter	Post. Mean	Post. SD	95% HDI
B	β_{B0} (intercept)	-0.32	0.37	[-1.08, 0.37]
	β_{B1} (age)	-0.65	0.37	[-1.38, -0.05]
	β_{B2} (gender)	-0.93	0.67	[-2.33, 0.34]
	β_{B3} (ASD diagnosis)	1.88	1.08	[-0.09, 4.10]
	β_{B4} (ID diagnosis)	-2.72	3.65	[-10.38, 3.62]
	β_{B5} (other diagnoses)	-5.39	3.23	[-11.97, -0.19]
C	β_{C0} (intercept)	-1.45	0.44	[-2.29, -0.58]
	β_{C1} (age)	0.24	0.33	[-0.39, 0.91]
	β_{C2} (gender)	0.02	0.70	[-1.32, 1.38]
	β_{C3} (ASD diagnosis)	2.88	0.93	[1.07, 4.71]
	β_{C4} (ID diagnosis)	2.27	1.75	[-0.92, 5.75]
	β_{C5} (other diagnoses)	-0.55	1.07	[-2.52, 1.61]
D	β_{D0} (intercept)	-1.60	0.33	[-2.26, -0.94]
	β_{D1} (age)	-0.12	0.27	[-0.65, 0.41]
	β_{D2} (gender)	0.02	0.60	[-1.16, 1.19]
	β_{D3} (ASD diagnosis)	3.40	0.85	[1.81, 5.09]
	β_{D4} (ID diagnosis)	0.53	1.79	[-2.75, 4.20]
	β_{D5} (other diagnoses)	-2.05	0.94	[-2.90, 0.77]

TABLE S8

Summary table containing the posterior mean estimated from MCMC samples for the CSHQ data using only item selection, along with the 95% HDI and posterior SD. The gender variable was binary encoded, taking value 1 for a response of female.

Group	Parameter	Post. Mean	Post. SD	95% HDI
B	β_{B0} (intercept)	-0.60	0.24	[-1.06, -0.16]
	β_{B3} (ASD diagnosis)	0.82	0.89	[-0.98, 2.51]
C	β_{C0} (intercept)	-1.33	0.35	[-2.01, -0.63]
	β_{C3} (ASD diagnosis)	2.96	0.89	[1.22, 4.69]
D	β_{D0} (intercept)	-1.67	0.37	[-2.42, -0.97]
	β_{D3} (ASD diagnosis)	2.30	0.87	[1.39, 4.82]

TABLE S9

Summary table containing the posterior mean estimated from MCMC samples for the CSHQ data using full variable selection, along with the 95% HDI and posterior SD. Estimates are computed over samples taken from iterations when the ASD diagnosis variable was included in the model.

S2.3. Variable Selection Results. Table S10 contains posterior inclusion probabilities for each item variable in the CSHQ survey.

CSHQ Subscale	Variable	PIP (LCR - Item)	PIP (LCR - simul.)	PIP (LCA)
Bedtime Resistance	1	0.00	0.00	0.00
	2	0.96	0.99	1.00
	3	1.00	1.00	1.00
	5	1.00	1.00	1.00
Sleep Onset Delay		1.00	1.00	1.00
Sleep Duration	1	1.00	1.00	1.00
	2	1.00	1.00	1.00
	3	1.00	1.00	1.00
Sleep Anxiety	1	1.00	1.00	1.00
	2	0.99	1.00	1.00
	3	1.00	1.00	1.00
	4	1.00	1.00	1.00
Night Waking	1	1.00	1.00	1.00
	2	1.00	1.00	1.00
	3	1.00	1.00	1.00
Parasomnias	1	0.00	0.00	0.00
	2	0.50	0.72	0.91
	3	0.99	1.00	1.00
	4	0.02	0.02	0.00
	5	1.00	1.00	1.00
	6	0.28	0.27	0.45
	7	0.00	0.00	0.00
Sleep Disordered Breathing	1	0.01	0.00	0.00
	2	0.08	0.09	0.01
	3	0.13	0.08	0.01
Daytime Sleepiness	1	0.15	0.19	0.12
	2	1.00	1.00	1.00
	3	0.48	0.42	0.29
	4	1.00	1.00	1.00
	5	1.00	1.00	1.00
	6	1.00	1.00	1.00
	7	0.00	0.00	0.00
	8	0.01	0.00	0.00

TABLE S10

Table containing the posterior inclusion probabilities (PIP) for each of the CSHQ items, from the item selection model for latent class regression (LCR - Item), the simultaneous item variable and predictor variable selection (LCR - simul.) and from the collapsed latent class analysis model (LCA)

S2.4. Supplementary Visualisations. Figures S3 and S4 contain mosaic plots for the each of the item variables of the CSHQ sleep patterns dataset which were retained by the collapsed sampler based on the posterior inclusion probabilities being greater than 0.5.

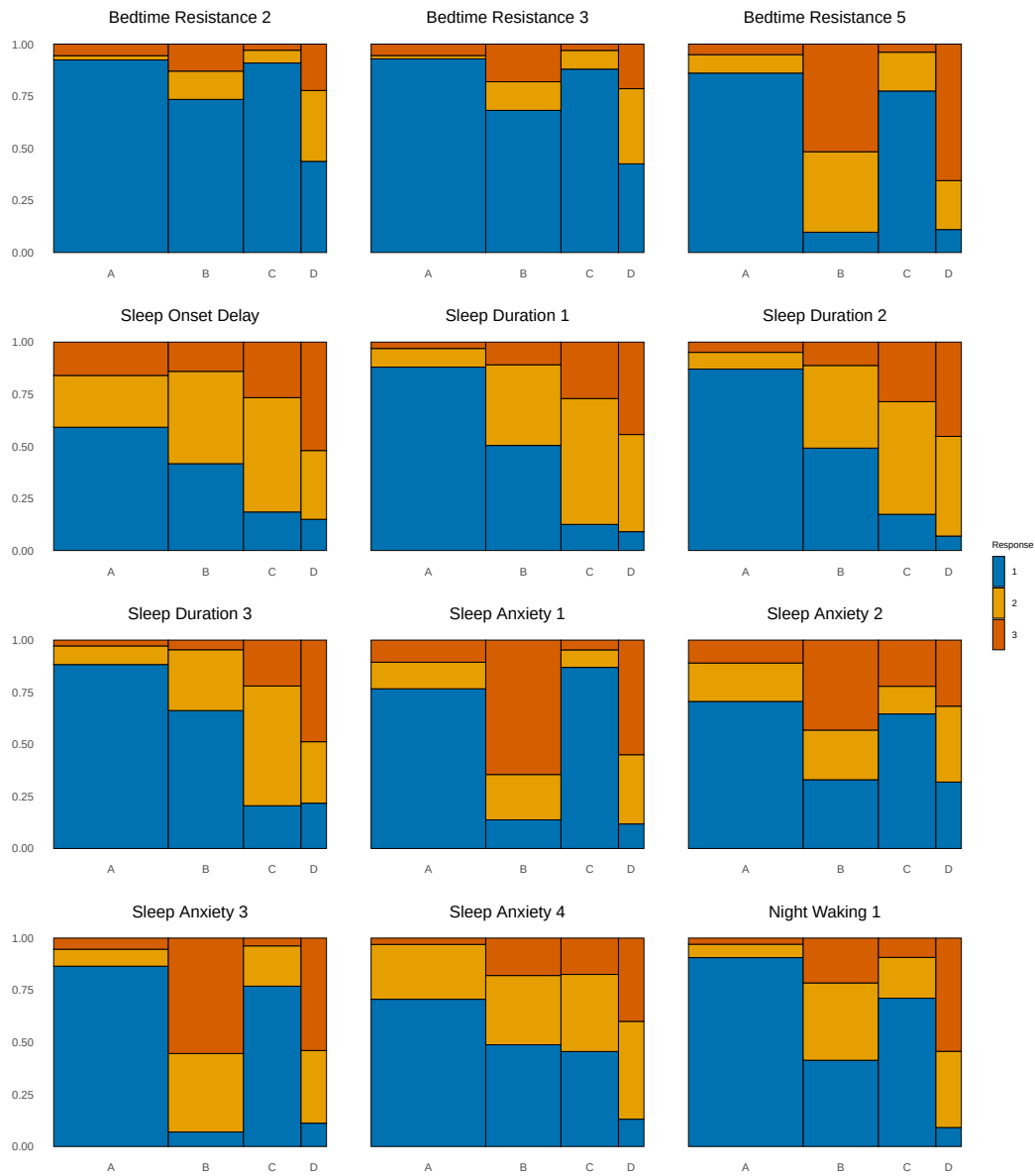


FIG S3.

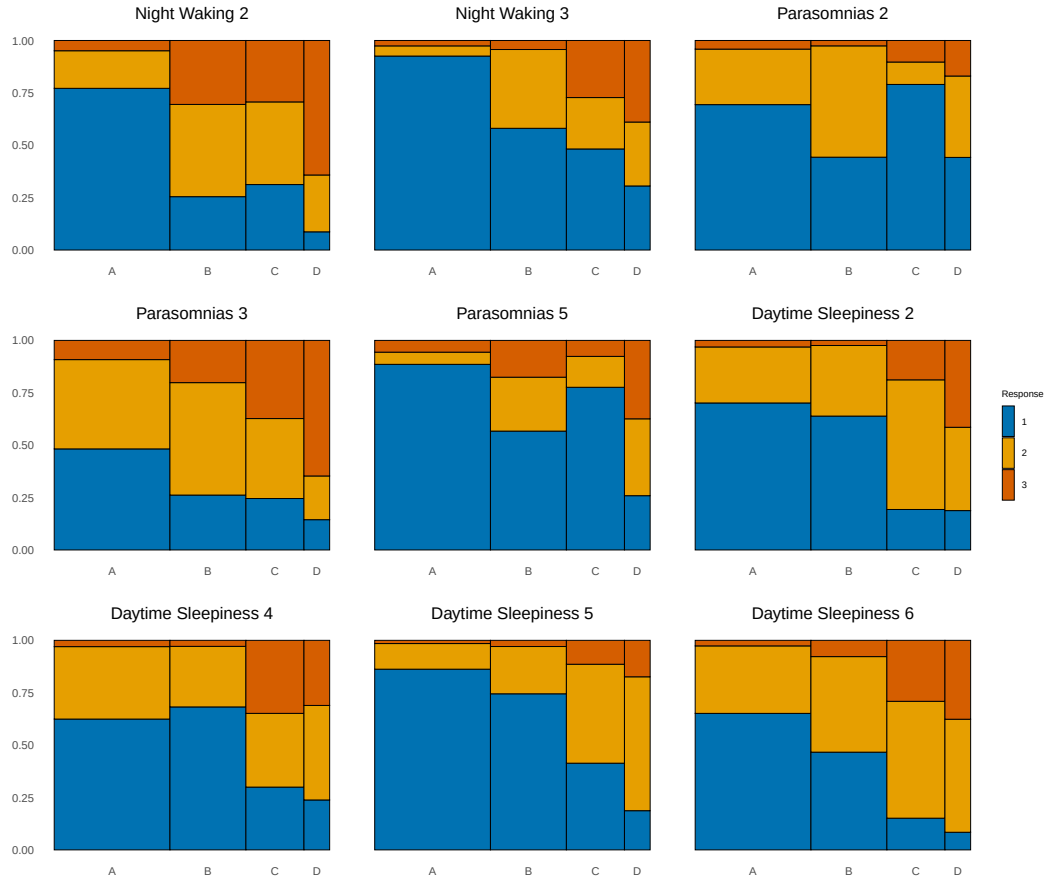


FIG S4.

S3. Technical Derivations.

S3.1. Derivation of Pólya-Gamma Sampling Method for LCR. This follows the method of Polson, Scott and Windle (2013). Consider the full conditional for β from (4). We can consider each of the β_g parameters conditional on β_{-g} , the β matrix with the g -th column removed. We write the expression for the full conditional distribution of β_g as (S1)

$$p(\beta_g | \beta_{-g}, \mathbf{X}, \mathbf{Y}) \propto p(\beta_g) \prod_{i=1}^n \left[\frac{\exp(\mathbf{X}_i^T \beta_g)}{\sum_{h=1}^G \exp(\mathbf{X}_i^T \beta_h)} \right]^{z_{ig}} \left[1 - \frac{\exp(\mathbf{X}_i^T \beta_g)}{\sum_{h=1}^G \exp(\mathbf{X}_i^T \beta_h)} \right]^{1-z_{ig}}.$$

This can be rewritten in the form

$$p(\beta_g | \beta_{-g}, \mathbf{X}, \mathbf{Z}) \propto p(\beta_g) \prod_{i=1}^n \frac{\exp(\eta_{ig})^{z_{ig}}}{1 + \exp(\eta_{ig})},$$

where

$$\eta_{ig} = \mathbf{X}_i^T \beta_g - C_{ig}, \quad C_{ig} = \log \left(\sum_{h \neq g} \exp(\mathbf{X}_i^T \beta_h) \right).$$

By utilising the result from (5), we can introduce the Pólya-Gamma random variables $\omega_{ig} \sim \text{PG}(1, 0)$ for each $i = 1, \dots, n, g = 1, \dots, G - 1$, which allows us to rewrite the full conditional as

$$p(\boldsymbol{\beta}_g | \boldsymbol{\beta}_{-g}, \mathbf{X}, \mathbf{Y}) \propto p(\boldsymbol{\beta}_g) \prod_{i=1}^n \exp \left(\kappa_{ig} \eta_{ig} - \frac{\omega_{ig} \eta_{ig}^2}{2} \right),$$

where $\kappa_{ig} = z_{ig} - \frac{1}{2}$. We can recognise the second multiplicative term here as a product of Gaussian kernels in the η variables. By setting the prior $\boldsymbol{\beta}_g \sim \mathcal{N}(\mathbf{m}_{0g}, \mathbf{V}_{0g})$, we are left with a two part update for $\boldsymbol{\beta}_g$ and ω_{ig} , given by

$$\begin{aligned} \boldsymbol{\beta}_g | \boldsymbol{\beta}_{-g}, \omega_{\cdot g}, \mathbf{X}, \mathbf{Z} &\sim \mathcal{N}(\mathbf{m}_g, \mathbf{V}_g), \\ \omega_{ig} | \boldsymbol{\beta}, \mathbf{X}_i &\sim \text{PG}(1, \eta_{ig}). \end{aligned} \tag{S2}$$

where

$$\mathbf{V}_g^{-1} = \mathbf{X}^T \Omega_g \mathbf{X} + \mathbf{V}_{0g}^{-1}, \quad \mathbf{m}_g = \mathbf{V}_g \left(\mathbf{X}^T (\boldsymbol{\kappa}_g + \Omega_g \mathbf{C}_{\cdot g}) + \mathbf{V}_{0g}^{-1} \mathbf{m}_{0g} \right), \quad \Omega_g = \text{diag}(\{\omega_{ig}\}_{i=1}^n),$$

and $\mathbf{C}_{\cdot g}$ is the g -th column of $\mathbf{C} = (C_{ig})$.

REFERENCES

POLSON, N. G., SCOTT, J. G. and WINDLE, J. (2013). Bayesian Inference for Logistic Models Using Pólya–Gamma Latent Variables. *Journal of the American Statistical Association* **108** 1339–1349. Publisher: [American Statistical Association, Taylor & Francis, Ltd.].